B

5K landscape | Growth × robustness | seed 42

gsMOBO family · exact ΔHV

gsMOBO family · total runtime

NSGA-II family · exact ΔHV

NSGA-II family · total runtime

S. cerevisiae

P. putida

C. glutamicum

CHO

E. coli

Exact ΔHV

Runtime (min)

Logical exact FBA calls

Algorithm

gsMOBO + landscape
gsMOBO + GHOST
gsMOBO + SPARC
gsMOBO + TRAIL

NSGA-II + landscape
NSGA-II + GHOST
NSGA-II + SPARC
NSGA-II + TRAIL

Landscape
GHOST
SPARC
TRAIL